PHIL1002: Introduction to Critical Thinking

Lecture 2A: Recognising Argument–II

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Agenda: Recognising Argument

Part 1
1. Recognising argument
2. Factors that can make it difficult to recognise an argument

Part 2
1. Factors that can make it look like there is an argument when there isn't.
2. Modal expressions

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Barriers to recognising argument

1. Distinction between the form of a sentence and its function
2. Inference indicators can be misleading
3. Nouns and noun phrases
4. Missing/implied conclusions

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1. Quotations and Descriptions

Factors that can make it look like there is an argument when there isn't.

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An example…

The great seventeenth-century French philosopher Descartes argued that, since it is possible for me to doubt the existence of my body (for an Evil Demon might be deceiving me into thinking I had a body when in fact I didn't), but it is not possible for me to doubt my own existence (for if I am doubting anything, I must exist to be doing the doubting), I cannot, therefore, be a body but must be something non-bodily.

Here the author is merely telling us what Descartes said, not arguing that Descartes was right when he said it.

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Compare this…

As the great seventeenth-century French philosopher Descartes argued, since it is possible for me to doubt the existence of my body (for an Evil Demon might be deceiving me into thinking I had a body when in fact I didn't), but it is not possible for me to doubt my own existence (for if I am doubting anything, I must exist to be doing the doubting), I cannot, therefore, be a body but must be something non-bodily.

By simply adding a word at the start of the passage, and dropping the word 'that,' I've converted the passage from one which *doesn't* argue a case into one which does.

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2. Explanations vs reasoning

Factors that can make it look like there is an argument when there isn't.

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Explanations vs reasoning

Since John suffered brain damage, he finds fine control of his movements impossible.
What does this sentence say?

1. ***From the time that*** *John suffered brain damage he has found fine control of his movements impossible.*
Or
2. ***Because*** *John suffered brain damage, he finds fine control of his movements impossible.*

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Explanations vs reasoning

(i) John finds fine control of his movements impossible because he suffered brain damage.
And
(ii) John has suffered brain damage; so, he will find fine control of his movements impossible.

(i) involves explanation of an acknowledged fact;
(ii) justification of a conclusion which the argument is meant to establish.

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Explanations vs reasoning

The distinction is genuine, and often the decision between explanation and argument is easy.

For example, this sentence:

The cause of John's lack of fine control of his movements is brain damage.

doesn't even look like reasoning, but it clearly offers and explanation, like this:

John lacks fine control of his movements. This is explained by the fact that he has suffered brain damage.

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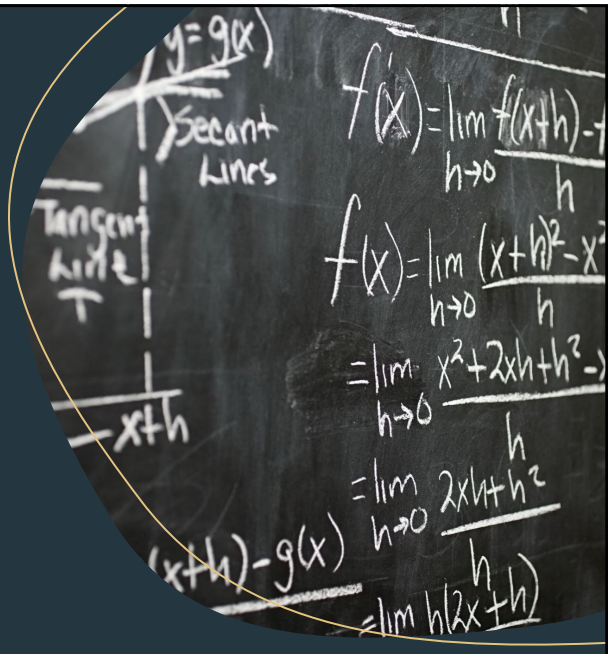
3. Conditional sentences

Factors that can make it look like there is an argument when there isn't.



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Conditional sentences

Conditional sentences have the form:

If X, then Y.

Where the X and Y indicate where statements go, as in this example:

'If he is at home, then the light will be on.'

Conditional sentences are NOT arguments.

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Why is a conditional sentence NOT an argument?

In an **argument**, a justification is offered for a conclusion, and both justification and conclusion are asserted (declared to be true)

Whereas, in a **conditional**, only the **conditional as a whole** is asserted, not its components.

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Conditionals…

Remember compound statements?

Conjunction and disjunction, which use 'and' and 'or' to make the compounds.

Conditionals are a third way of forming a compound statement, by connecting two simple statements with the words 'if' and 'then'.

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Conditionals…

The general form of a conditional is:

If (antecedent), then (consequent).

Thus, in our example the antecedent is *'he is at home'* and the consequent is *'the light will be on'*.

In everyday English the word 'then' is often omitted, and so is the comma, as in:

If he's at home the light will be on.

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Conjunctions..

Remember how we can also form compound statements by conjunction, where the general form is: **X & Y.**
E.g. *Grandad cooks the children's pets and Grannie eats them.*
Someone who declares this conjunction to be true, is asserting both
Conjunct 1: *Grandad cooks the pets.* **AND**
Conjunct 2: *Grannie eats the pets.*

For the whole conjunction to be true, each of the conjuncts must be true independently.

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Conjunctions are *similar* to arguments…

Someone who says:

'Grandad cooks the pets, ***so*** *Grannie eats them'*

is also asserting that **both of the statements which make up the argument are true.**

BUT she is doing more than this, of course: she is saying that **there is a connection between them.**

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NOT SO for conditionals

Conjunctions and arguments are different in this respect from conditionals.
What is asserted by someone who says this?

If Grandad cooks the pets, then Grannie eats them.

This does **NOT** say that Grannie does eat the pets, nor that Grandad cooks them.
The word **'if'** cancels the assertion of the components; **only the conditional as a whole is asserted.**

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Conditionals

Conditionals allow us to consider situations when we don't know whether they exist or not, or even when we do know that they don't exist, e.g.:

If the Vice-Chancellor is not in his office, he is at the Uni Club.

If Germany had not invaded the Soviet Union, then she would have won the second World War.

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Conditionals are NOT arguments

In conditionals, the two components are *indivisible* as far as assertion goes—you can't detach the consequent and maintain it on its own.

But in arguments, the components are separately asserted, and the conclusion **can** be detached and maintained on its own.

Single conditionals are NOT arguments.

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Summary: factors that can make it look like there is an argument when there isn't.

1. Quotations and descriptions
2. Explanations vs reasoning/argument
3. Conditional sentences

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Agenda: Recognising Argument

Part 1
1. Recognising argument
2. Factors that can make it difficult to recognise an argument

Part 2
1. Factors that can make it look like there is an argument when there isn't.
2. **Modal expressions**

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Modal expressions

Modal expressions are things like:

'must' 'necessarily' 'possibly' 'may' 'can' 'must not' 'cannot' 'may not' 'probably'

and so on.

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Modal expressions are used...

1. To say that certain statements are necessarily true or necessarily false.
2. To make statements concerning capacities and opportunities.
3. To act as inference indicators.

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1. To express necessary truth.

A very young dog is a puppy

*A human being **CANNOT BE** in two places at the same time*

*Different colours **NECESSARILY** exclude each other from the same area*

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Necessary truth…

Here's another example of a necessary truth:
3 + 4 = 7

NECESSARILY, 3 + 4 = 7

3 and 4 MUST equal 7

3 and 4 HAVE TO BE 7

The sum of 3 and 4 CANNOT be other than 7

The sum of 3 and 4 CAN'T be other than 7

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Necessity and contingency

Necessary truth: a true statement which couldn't possibly have been false.

Contingent truth: a true statement which is, as a matter of fact, true, but could possibly have been false.

Necessary falsehoods: statements that **couldn't possibly** have been true.

Contingent falsehoods: statements which aren't true but which **might have been**.

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Contingent truths...

English is the international language of air traffic control.

The Vice-Chancellor of The University of Western Australia is male.

There wasn't an earthquake in Perth yesterday.

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Necessary falsehoods

17 is greater than 21.

Fish are mammals.

Biological parents are younger than their children.

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Contingent falsehoods

Apples are not grown in Tasmania.

Australia's smallest coin is now the $5 piece.

Mozart composed his best music after the age of 50.

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2. Statements concerning capacities and opportunities.

- *It is both POSSIBLE and NECESSARY for you to attend tutorials.*
- *Suzie CAN walk well in high heels but Jane CAN'T.*
- *Murgatroyd COULD HAVE come to the meeting.*
- *It's IMPOSSIBLE for me to reach that shelf.*
- *Aqua regia CAN dissolve gold.*
- *An untrained human being CANNOT jump a two-metre bar.*
- *If you want to keep within the law, you MUST NOT drive at more than 40 km/h in King's Park.*
- *I HAVE TO go to a meeting tomorrow.*

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3. Inference indicators

John <u>must be</u> at home: I hear someone breathing behind the door.

What I have in my hand <u>must be</u> a diamond, because I have just scratched my name on the window with it.

What I have in my hand <u>cannot be</u> plastic, because I have just scratched my name on the window with it.

Although Henry killed the other girls, he <u>can't</u> have killed Jane, <u>because</u> someone else's fingerprints are on the murder weapon.

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Modal inference indicators

*I hear someone breathing behind the door; therefore, John **is** at home.*

*I have just scratched my name on the window with what I have in my hand; therefore, it **is** a diamond.*

*I have just scratched my name on the window with what I have in my hand; therefore, it **is not** plastic.*

*Someone else's fingerprints are on the murder weapon; therefore, Henry **didn't** kill Jane.*

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Relative necessity

When modal expressions occur as inference indicators, they are NOT there to suggest that the conclusions are *absolutely* necessary truths.

They indicate that the conclusions are *relatively necessary*, that is, they are necessary relative to the premises—they are necessitated **by** the premises.

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Modal inference indicators

Modal words can do the job of reinforcing the conclusion, and indicating that, given the premises are true the conclusion must be true.

They emphasise that, given one contingent truth, another truth, equally contingent in its own right, follows logically from the first one.

That doesn't mean that the conclusion is necessarily true, in its own right, regardless of the truth of the premise.

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Modal confusion...

The potential for confusion is enormous when one starts reasoning **about** necessities and contingencies or about capacities and opportunities.

This is because arguments about these matters are likely to involve modal words in their conclusions.

So, it may not be clear which of the three functions these words are performing.

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Modal confusion

If you know something, you can't be wrong about it (by the definition of 'know').

But you can always be wrong about anything (after all, we sometimes make mistakes even about very obvious things).

So, you can never know anything.

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An analogous argument..

If you've kicked a goal, you can't have missed (by the definition of 'kicking a goal').

But you could always have missed (after all, even very good players sometimes miss easy goals).

So, you can never kick a goal.

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Summary

1. Although the conclusion of a valid argument can be a contingent truth, within the context of the argument it may be expressed using a modal phrase; and we can then mistake the conclusion for a necessary truth when really all the modal phrase is doing is acting as an inference indicator.

2. In reasoning about freewill, we are likely to want conclusions about capacities for action. One may then mistake an inferential use of a modal phrase for a conclusion attributing capacity or lack of capacity.

3. In some arguments, a modal phrase may be doing double duty, e.g. ***both*** acting as an inference indicator ***and*** attributing necessity (or capacity or probability) or lack of it. And it may be impossible to be sure whether both duties are being done or not.

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The moral of the story...

Watch out for modal expressions!

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Summary

Factors which can make you overlook the presence of reasoning:

1. Distinction between the form of a sentence and its function
2. Inference indicators can be misleading
3. Nouns and noun phrases
4. Missing/implied conclusions

Factors which can make you mistakenly suppose that reasoning is present when it isn't:

1. Quotation and descriptions
2. Explanation vs reasoning/argument
3. Conditional sentences

Modal expressions!

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Next lecture....

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